

API Vulnerability Detection in Open Banking Ecosystems

Margaret Cullet
University of Bristol

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Abstract

The proliferation of open banking ecosystems has fundamentally transformed financial services, introducing unprecedented connectivity and innovation while simultaneously expanding the attack surface through widespread API adoption. As financial institutions increasingly rely on APIs for open banking, payment processing, and real-time data interchange, the risk landscape has grown more complex and consequential, with API-related breaches becoming a critical concern for the sector.

This research examines the limitations of traditional security approaches—including static application security testing, dynamic application security testing, and conventional web application firewalls—which have proven inadequate for addressing the unique challenges of API security in open banking contexts. These conventional tools struggle with identifying business logic abuse, detecting zero-day exploits, and managing the scale of modern API deployments. Drawing on established security frameworks and vulnerability databases, this research identifies critical vulnerability patterns including broken object-level authorization, improper access controls, and injection attacks that disproportionately affect financial APIs.

This paper proposes a comprehensive detection framework integrating machine learning-driven behavioral analytics with runtime protection mechanisms. The framework leverages anomaly detection to identify suspicious transaction patterns, continuous API schema learning for proactive threat identification, and automated threat intelligence integration to predict and mitigate emerging cyber threats. Additionally, the research examines the implementation of Financial-Grade API security standards, mutual TLS authentication, and certificate validation protocols mandated by open banking regulations.

By synthesizing vulnerability trend analysis with practical security controls across the API software development lifecycle, this study contributes a cybersecurity resilience model specifically tailored for financial institutions, enabling enhanced detection capabilities, reduced mean time to mitigation, and strengthened regulatory compliance within open banking ecosystems.

Keywords: Open Banking; API Vulnerability Detection; Cybersecurity; Machine Learning; Threat Intelligence; Financial-Grade APIs; Regulatory Compliance; Anomaly Detection

1. Introduction

1.1 Background

The financial services industry is undergoing a profound transformation driven by the convergence of digital innovation, regulatory change, and evolving consumer expectations. At the heart of this transformation lies open banking—a paradigm that mandates financial institutions to share customer-permissioned data with licensed third-party providers through standardized Application Programming Interfaces (APIs). This regulatory-driven shift, exemplified by the European Union's Revised Payment Services Directive (PSD2), the United Kingdom's Open Banking Standard, and the United States' Consumer Financial Protection Bureau Section 1033 rule, has fundamentally redefined how financial data is accessed, shared, and utilized (Broby, 2021).

APIs serve as the connective tissue of this new financial ecosystem, enabling seamless integration between banks, fintech innovators, payment processors, and consumer applications. According to industry research conducted by the Financial Services Information Sharing and Analysis Center (FS-ISAC, 2023), more than 85% of banks now participate in open banking ecosystems, and average API call volumes in the financial sector doubled from 2023 to 2024. This connectivity has unleashed unprecedented innovation, allowing consumers to aggregate accounts across multiple institutions, access personalized financial management tools, and benefit from competitive lending and payment services.

However, the same attributes that make open banking transformative—openness, connectivity, and data sharing—have simultaneously created an expanded and highly attractive attack surface for cybercriminals. APIs, by their nature, expose critical systems and sensitive financial data to external parties, making them prime targets for exploitation (Coventry and Branley, 2018). The security implications are stark: recent surveys indicate that 57% of financial services firms reported experiencing API-related breaches in the past two years, and a staggering 88.7% experienced an API-related security incident, representing the highest rate across all industry sectors (Salt Security, 2024).

1.2 Problem Statement

The security challenges facing open banking ecosystems are multifaceted and evolving rapidly. Traditional security approaches, including Static Application Security Testing (SAST), Dynamic Application Security Testing (DAST), and conventional Web Application Firewalls (WAFs), have proven inadequate for addressing the unique demands of API security in financial contexts (Sillaber et al., 2021). These legacy tools generate high volumes of false positives, require extensive manual validation, and cannot operate at the speed or depth required by modern financial institutions. Moreover, they are fundamentally ill-suited to detect business logic abuses—one of the most dangerous categories of API vulnerabilities—because they lack contextual understanding of intended transaction flows and user behaviors (Zeller, 2021).

The problem is compounded by several factors unique to open banking. First, the sheer volume and velocity of API transactions in financial environments make manual security review impractical. Financial institutions process billions of API calls daily, with peak transaction rates reaching tens of thousands per second (Accenture, 2023). Second, the complexity of

modern API architectures—incorporating microservices, containerization, and multi-cloud deployments—creates numerous points of potential failure that traditional perimeter-based security models cannot adequately protect (Rosa and Garriga, 2020). Third, the regulatory landscape imposes stringent security requirements while simultaneously mandating openness, creating tension between compliance obligations and threat mitigation strategies (Alhassan and Othman, 2023).

Third-party risk further complicates the threat landscape. Open banking ecosystems involve extensive sharing of sensitive financial data with external providers—including account aggregation services, payment initiation providers, and financial management applications—each of which represents a potential vector for compromise (Ryu, 2023). The interconnected nature of these systems means that a vulnerability in a single third-party API can cascade across the entire ecosystem, potentially exposing millions of customer accounts and billions in assets (Huang and Wang, 2022).

1.3 Research Objectives

This research addresses the critical need for effective API vulnerability detection in open banking ecosystems through the following objectives:

Objective One: To systematically analyze the current threat landscape for APIs in open banking environments, identifying the most prevalent and consequential vulnerability categories based on empirical data and industry incident reports.

Objective Two: To evaluate the limitations of existing security approaches—including SAST, DAST, and WAFs—specifically within the context of financial APIs, and to identify the gaps that attackers are exploiting.

Objective Three: To develop a comprehensive detection framework that integrates machine learning-driven behavioral analytics with runtime protection mechanisms, designed specifically for the unique requirements of open banking environments.

Objective Four: To propose a cybersecurity resilience model tailored for financial institutions that enables enhanced detection capabilities, reduced mean time to mitigation, and strengthened regulatory compliance.

1.4 Research Questions

This study is guided by the following research questions:

RQ1: What are the most critical API vulnerability categories affecting open banking ecosystems, and what patterns characterize attacks exploiting these vulnerabilities?

RQ2: Why do traditional security approaches fail to adequately detect and prevent API-based attacks in financial contexts?

RQ3: How can machine learning and behavioral analytics be effectively applied to detect anomalous API transactions indicative of malicious activity?

RQ4: What security controls and architectural practices are most effective in strengthening API resilience within open banking environments?

1.5 Scope and Delimitations

This research focuses specifically on API security within open banking ecosystems, with particular emphasis on financial institutions operating under PSD2 and UK Open Banking regulatory frameworks. The study encompasses both external-facing public APIs and partner APIs used for third-party integration, but does not address internal APIs that are not exposed outside the organization. The geographical scope is primarily Europe and the United Kingdom, given their leadership in open banking adoption, although relevant comparisons with other jurisdictions—including Australia, Canada, and the United States—are included where appropriate.

The research excludes non-API attack vectors such as phishing, social engineering, and insider threats, except where these intersect with API vulnerabilities. The study also does not address cloud infrastructure security more broadly, except where directly relevant to API deployment. The proposed detection framework is conceptual rather than implemented, though it draws extensively on proven techniques from existing implementations (Singh and Sharma, 2023; Feng et al., 2021).

1.6 Significance of the Study

This research makes several significant contributions to both academic knowledge and industry practice. From a theoretical perspective, it advances understanding of API security vulnerabilities in a unique and increasingly important context—the highly regulated, data-rich environment of open banking. While extensive literature exists on API security generally (Tandon et al., 2023; Li et al., 2022), limited research has specifically addressed the intersection of API vulnerabilities, financial services, and regulatory compliance.

From a practical standpoint, the detection framework proposed in this research provides financial institutions with actionable guidance for strengthening their API security posture. The framework addresses the limitations of current approaches while being grounded in the operational realities of financial institutions—including regulatory obligations, performance requirements, and risk tolerance (Borse and Mankar, 2023). The proposed resilience model offers a structured approach to evaluating and improving API security capabilities, potentially reducing both the likelihood and impact of API-related breaches.

The research also contributes to policy discussions by highlighting areas where regulatory requirements may create security tensions and by identifying security controls that could be more explicitly mandated in open banking standards. As open banking expands globally—with emerging frameworks in Australia, Brazil, Canada, and beyond—the findings of this research will be relevant to policymakers and regulators seeking to establish effective security requirements (BIS, 2022).

1.7 Thesis Structure

The remainder of this thesis is organized as follows:

Section 2: Literature Review provides a comprehensive examination of existing research relevant to API vulnerability detection in open banking ecosystems. It covers the open banking landscape, API security fundamentals, the OWASP API Security Top 10, machine learning applications in security, and regulatory frameworks.

Section 3: Methodology details the research approach, including the systematic literature review methodology, case study selection criteria, framework design approach, and ethical considerations.

Section 4: Results presents the findings from the case study analysis and vulnerability trend analysis, providing both qualitative and quantitative insights into the current state of API security in open banking.

Section 5: Discussion of Findings interprets the results in light of existing literature, addresses the research questions, and presents the proposed detection framework and resilience model.

Section 6: Conclusion summarizes the key findings, discusses implications for practice and policy, acknowledges limitations, and suggests directions for future research.

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2. Literature Review

2.1 The Open Banking Landscape

2.1.1 Evolution and Regulatory Foundations

The concept of open banking emerged from a confluence of technological advancement, consumer advocacy, and regulatory intervention aimed at fostering competition and innovation in financial services (Brown and Pike, 2022). The regulatory foundations were laid in Europe with the publication of PSD2 in 2015, which mandated that banks provide third-party providers with access to customer account data through secure APIs, subject to explicit customer consent (European Commission, 2015). The UK followed with its Open Banking Standard in 2018, establishing a common API standard for account information and payment initiation services (OBIE, 2018).

The rationale behind these regulatory initiatives was compelling. Prior to open banking, customer financial data was siloed within individual institutions, creating switching costs and inhibiting competition (Zachariadis and Ozcan, 2017). Consumers could not easily view their complete financial picture across multiple accounts, compare product offerings effectively, or benefit from personalized financial advice based on comprehensive data. The open banking approach—mandating data portability through standardized APIs—was designed to address these market failures, empower consumers, and stimulate innovation (Babichenko, 2023).

Following the European and UK initiatives, open banking frameworks have been adopted or are in development globally. Australia implemented its Consumer Data Right in 2020, beginning with banking and expanding to energy and telecommunications (ACCC, 2020). Brazil introduced its open banking framework in 2021, with phased implementation of data sharing and service delivery (BCB, 2021). Canada is developing a consumer-driven banking framework, while the United States is advancing rules for financial data sharing under the Consumer Financial Protection Bureau's Section 1033 of the Dodd-Frank Act (CFPB, 2023).

2.1.2 Open Banking Architecture and API Usage

The technical architecture of open banking is fundamentally API-centric, comprising several key components and interactions (He et al., 2022). At its core, the architecture includes:

API Gateways serve as the entry point for all external API requests, providing authentication, rate limiting, and request routing. In open banking environments, API gateways must support OAuth 2.0 and OpenID Connect (OIDC) for authorization, as mandated by the Financial-grade API (FAPI) security profile (FAPI WG, 2022). The gateway must also handle the unique requirements of PSD2, including Strong Customer Authentication (SCA) and dynamic client registration.

Account Information Service Providers (AISPs) access customer account data from banks through the API, enabling account aggregation, financial management, and data analytics services. These third-party providers must be licensed or registered with relevant authorities and must comply with data protection and security requirements (EBA, 2022).

Payment Initiation Service Providers (PISPs) initiate payments directly from customer bank accounts through the API, eliminating the need for card-based transactions. PISPs require elevated security controls given their ability to execute financial transactions (Goldstein et al., 2020).

Account Servicing Payment Service Providers (ASPSPs)—the banks themselves—expose APIs to AISPs and PISPs while maintaining responsibility for customer account security and transaction integrity.

The API landscape in open banking is remarkably dynamic. Research indicates that the average financial institution now manages over 1,200 distinct APIs, with half of these exposed externally (Postman, 2023). API call volumes have grown exponentially, with leading banks processing billions of API transactions monthly (Finastra, 2024). This scale, combined with the sensitive nature of financial data, makes robust API security an operational imperative.

2.1.3 Security Requirements and Challenges

The security requirements for open banking APIs are exceptionally demanding, reflecting the high value of financial data and the potential for systemic risk (Chen and Wu, 2022). At minimum, APIs must provide:

Authentication and Authorization: FAPI compliance requires the use of OAuth 2.0 with Proof Key for Code Exchange (PKCE) and mTLS for client authentication. OIDC is used for identity verification, and all tokens must be cryptographically signed and encrypted (FAPI WG, 2022).

Data Protection: API payloads containing personal and financial data must be protected in transit using TLS 1.2 or higher, and sensitive data elements may require additional encryption or tokenization (MacKenzie and Smith, 2023). Consent management is crucial, as APIs must ensure that third-party providers only access data explicitly authorized by the customer.

Transaction Integrity: APIs must implement mechanisms to verify the integrity of transactions and detect tampering or replay attacks. This includes the use of digital signatures, nonces, and timestamps (NIST, 2020).

Availability and Performance: Financial APIs must meet stringent availability requirements, often exceeding 99.99% uptime. Denial of service protection is essential, as service disruptions can have severe financial and reputational consequences (Tan and Lee, 2021).

Despite these requirements, implementing effective security in practice presents significant challenges. The complexity of modern API deployments—spanning multiple clouds, legacy systems, and external partners—creates numerous potential attack vectors (Veracode, 2023). Regulatory fragmentation across jurisdictions complicates compliance, while the pace of innovation often outstrips security controls (Weber and Shin, 2022).

2.2 API Security Fundamentals

2.2.1 The Nature of API Vulnerabilities

API vulnerabilities differ fundamentally from traditional web application vulnerabilities, primarily because APIs are designed for machine-to-machine communication rather than human interaction (Sood and Enbody, 2021). This distinction has profound security implications. Web applications typically present a user interface that constrains possible interactions, whereas APIs expose programmatic interfaces that can be accessed in arbitrary ways by sophisticated attackers.

The unique characteristics of API vulnerabilities can be understood through several dimensions. First, APIs expose structured data and functionality in ways that facilitate automated attack at scale. An attacker can script interactions with an API to test hundreds of thousands of parameter combinations or attempt to traverse object identifiers sequentially (Sullivan and Choffnes, 2022). Second, APIs are often developed by different teams than the applications that consume them, leading to misalignments in security assumptions and implementation (Sacco and Ruggiero, 2023).

Third, APIs inherently expose more attack surface than web applications. Where a web application might expose dozens of endpoints, a modern API can expose hundreds or thousands, each representing potential vulnerabilities (Ahmed et al., 2022). Fourth, the complexity of API interactions—often involving chained calls across multiple services—

makes it difficult to understand the full security implications of changes or to test all possible states (Kotsiantis and Kanellopoulos, 2021).

2.2.2 The OWASP API Security Top 10

The Open Web Application Security Project (OWASP) has been instrumental in categorizing and raising awareness of API-specific vulnerabilities. The OWASP API Security Top 10, most recently updated in 2023, provides a framework for understanding the most critical API security risks (OWASP, 2023).

API1:2023 Broken Object Level Authorization (BOLA) represents the most prevalent and dangerous API vulnerability. BOLA occurs when an API endpoint does not properly verify that the requesting user is authorized to access the specific object they are requesting—for example, accessing another user's account details by modifying a numeric ID in the API request (Mansfield-Devine, 2021). Research suggests that BOLA vulnerabilities account for more than 30% of all API security incidents in financial services (Traceable AI, 2023).

API2:2023 Broken Authentication encompasses vulnerabilities in authentication mechanisms, including weak credentials, improper token handling, and session management failures. In open banking contexts, compromised authentication can enable attackers to impersonate legitimate third-party providers or customers (Khurana and Amin, 2023).

API3:2023 Broken Object Property Level Authorization occurs when APIs expose excessive data in responses or allow inappropriate modification of object properties. In open banking, this might involve an API returning sensitive fields—such as account balances or transaction histories—that should be withheld based on consent scope (Harrison and Wright, 2022).

API4:2023 Unrestricted Resource Consumption involves the lack of rate limiting or resource controls, enabling attackers to overwhelm API resources, cause denial of service, or bypass billing controls. Financial APIs are particularly vulnerable given the high value of requests such as payment initiation or credit scoring (Pratap and Singh, 2023).

API5:2023 Broken Function Level Authorization occurs when authorization policies are not properly enforced at the function level, allowing attackers to access privileged functionality. For example, an authenticated user might be able to invoke administrative functions by calling endpoints that lack proper role checking (Algarni and Jones, 2021).

API6:2023 Unrestricted Access to Sensitive Business Flows involves the ability to access sensitive business processes without authorization. In financial APIs, this might include accessing fraud detection algorithms, risk scoring models, or transaction monitoring systems (Frye and Black, 2023).

API7:2023 Server Side Request Forgery (SSRF) enables attackers to induce the API server to make requests to unintended destinations, potentially accessing internal systems or cloud metadata services. SSRF is particularly dangerous in cloud-native financial environments (Chen and Zhang, 2023).

API8:2023 Security Misconfiguration encompasses a broad range of configuration vulnerabilities, from unpatched software and exposed debug endpoints to misconfigured CORS policies. The complexity of modern API deployments makes misconfiguration a persistent challenge (White et al., 2022).

API9:2023 Improper Inventory Management involves failures in maintaining an accurate inventory of API endpoints, leading to outdated or shadow APIs—endpoints that exist outside the formal API inventory and may lack proper security controls. Research indicates that 40% of financial institutions have undiscovered shadow APIs in their environments (SecurityWeek, 2023).

API10:2023 Unsafe Consumption of APIs occurs when APIs consume data from other services without proper validation, enabling injection attacks or data poisoning (Brown and Johnson, 2023).

2.2.3 Financial-Specific API Vulnerabilities

While the OWASP Top 10 provides a comprehensive overview of general API vulnerabilities, financial APIs face additional risks specific to their context. Several studies have identified vulnerability patterns that are particularly relevant to open banking environments.

Business Logic Abuse represents one of the most significant threats to financial APIs. Business logic vulnerabilities occur when attackers exploit the intended functionality of an application to achieve a malicious outcome—for example, manipulating transaction sequences, exploiting race conditions in fund transfers, or bypassing payment validation flows (Tucker and Tobin, 2022). These vulnerabilities are notoriously difficult to detect with automated tools because they involve violations of expected behavior rather than coding errors.

Research by the European Banking Authority (EBA, 2023) identified business logic abuse as a growing concern in open banking, with incident reports increasing by 60% annually. Attackers have exploited vulnerabilities in payment initiation flows to redirect funds, manipulated consent management to extend access beyond authorized scopes, and exploited timing vulnerabilities in multi-factor authentication processes (Kundu and Mishra, 2022).

Financial Data Exfiltration is another critical concern. APIs handling financial data—including account balances, transaction histories, payment details, and credit information—are prime targets for data theft (Van der Merwe and Du Toit, 2022). Unlike other sectors where data breaches may primarily damage reputation, financial data breaches can enable immediate financial fraud, identity theft, and sophisticated social engineering attacks (Ponemon Institute, 2023).

Man-in-the-Middle (MITM) and Interception Attacks pose particular risks to financial APIs, especially where strong authentication and encryption are not consistently implemented (Tian and Wang, 2021). The high value of financial data makes it worth the effort required to compromise TLS connections, intercept tokens, or perform adversary-in-the-middle attacks on API communication.

API Injection Attacks including SQL injection, command injection, and LDAP injection, remain prevalent in financial APIs despite extensive awareness of the risks. A study of financial API vulnerabilities found that 45% of APIs contained some form of injection vulnerability, often in areas such as search functionality, parameterized queries, or dynamic query building (Checkmarx, 2023).

Zero-Day Exploitation is an increasingly prominent concern as attackers target previously unknown vulnerabilities in financial APIs. The growing sophistication of API threat actors—including state-sponsored groups and organized crime organizations—has led to an increase in zero-day exploitation in the financial sector (Mandiant, 2023).

2.3 Traditional Security Approaches and Their Limitations

2.3.1 Static Application Security Testing (SAST)

Static Application Security Testing analyzes source code, bytecode, or binary code for security vulnerabilities without executing the application. SAST tools examine code structure, data flow, and control flow to identify potential vulnerabilities such as buffer overflows, SQL injection, and cross-site scripting (Huang et al., 2022).

In the context of API security, SAST offers the advantage of early detection—vulnerabilities can be identified during development, before deployment. This shift-left approach aligns with DevSecOps principles and can reduce the cost of fixing vulnerabilities (Janssen and Johnson, 2023). However, SAST has significant limitations for API security in open banking environments.

First, SAST is inherently limited to analyzing code paths that can be determined statically. It cannot identify vulnerabilities that emerge from runtime behavior, such as business logic flaws, authentication bypasses that require stateful interactions, or vulnerabilities dependent on specific runtime configurations (Davis and Williams, 2022). Financial APIs often depend heavily on these runtime behaviors, making SAST insufficient on its own.

Second, SAST tools generate high volumes of false positives, particularly in complex codebases. Financial institutions report that as many as 70% of SAST findings are false positives, requiring significant manual effort to triage and validate (Gartner, 2023). This high false positive rate can lead to alert fatigue, where genuine vulnerabilities are overlooked amidst the noise.

Third, SAST tools typically struggle with modern API architectures, including microservices and asynchronous communication patterns. The distributed nature of microservices means that SAST tools cannot easily trace data flows across service boundaries, leading to incomplete analysis and missed vulnerabilities (Arora and Satish, 2022).

Fourth, SAST tools are language-dependent and require integration with development pipelines. For financial institutions with heterogeneous technology stacks—including legacy systems, multiple programming languages, and custom frameworks—comprehensive SAST coverage is challenging and resource-intensive (Powers and Kumar, 2023).

2.3.2 Dynamic Application Security Testing (DAST)

Dynamic Application Security Testing takes a different approach, analyzing a running application by simulating external attacks. DAST tools send crafted inputs to API endpoints and analyze responses for signs of vulnerabilities, including injection flaws, authentication issues, and data exposure (Vega and Rodriguez, 2022).

DAST offers the advantage of testing the API as it is actually deployed, with real configurations and dependencies. This can identify vulnerabilities that depend on runtime conditions and can validate the effectiveness of other security controls (Chen and Lee, 2022). However, DAST also has significant limitations for open banking APIs.

The most critical limitation is that DAST cannot exercise complex business logic flows. Financial APIs often involve multi-step transactions, state-dependent authorization, and conditional behaviors that DAST tools cannot simulate effectively. For example, a payment initiation flow involving authentication, consent validation, account verification, and transaction authorization requires specific sequences of API calls with valid credentials—DAST tools are not designed to navigate such complex flows (Martinez and Cooper, 2023).

DAST tools are also limited by their reliance on API documentation. To test an API effectively, DAST tools require knowledge of endpoints, parameter formats, and expected responses. If documentation is incomplete or inaccurate, DAST coverage will be correspondingly limited (Lee and Patel, 2022). In practice, many financial APIs maintain extensive documentation, but the dynamic nature of modern APIs means that documentation often lags behind implementation.

Additionally, DAST tools are limited by their inability to distinguish between different authorization contexts. While DAST can test authentication controls, it cannot effectively test for object-level authorization vulnerabilities—the most common and dangerous class of API vulnerabilities—because this requires understanding which objects a particular user should be able to access (Smith and Jones, 2022).

The performance impact of DAST testing is also a concern. Comprehensive testing of financial APIs requires thousands of requests, which can impact performance in shared environments and may interfere with production traffic if not carefully managed (Tiwari and Gupta, 2023).

2.3.3 Web Application Firewalls (WAFs)

Web Application Firewalls sit between API clients and servers, inspecting traffic and applying rules to block malicious requests. Modern WAFs can be deployed as hardware appliances, cloud services, or software agents, and increasingly incorporate machine learning for threat detection (Ellison and Foster, 2022).

For financial APIs, WAFs provide several important capabilities. They can block known attack patterns, including SQL injection and cross-site scripting. They can enforce rate limiting to prevent denial-of-service attacks. They can implement bot mitigation to filter automated

attacks. And they can provide visibility into API traffic, enabling security monitoring and incident response (Rashid and Malik, 2023).

However, WAFs have critical limitations in the context of open banking APIs. Traditional WAFs operate at the application layer and primarily protect against known attack signatures. They cannot detect business logic abuse, which involves legitimate-looking requests that violate intended application behavior (Borges and Oliveira, 2022). An attacker exploiting a race condition in a payment flow would not trigger any WAF rule because all requests are individually valid.

WAFs also cannot address authorization vulnerabilities such as BOLA. If an authenticated user requests another user's account balance by modifying a user ID parameter, the WAF sees a valid request to a valid endpoint with valid authentication—it has no way to know that the user is not authorized for that particular account (Weber and Gupta, 2023).

The effectiveness of WAFs against zero-day attacks is also limited. Signature-based WAFs require signatures for each attack pattern, which must be developed and deployed before they can provide protection. While modern WAFs incorporate behavioral analysis and anomaly detection, these capabilities are often immature and limited in their ability to detect sophisticated API attacks (Whitaker and Chen, 2022).

Furthermore, WAFs can introduce significant operational challenges. They must be carefully configured to avoid blocking legitimate traffic while catching malicious requests. In complex financial API environments with extensive integrations and varied use cases, finding the right balance is difficult. Overly aggressive blocking can disrupt legitimate services, while overly permissive configurations leave APIs vulnerable (Murphy and Walsh, 2022).

2.3.4 Automated Vulnerability Scanners

Automated vulnerability scanners represent a broader category of tools that combine aspects of SAST, DAST, and WAFs to identify vulnerabilities across the development lifecycle. These tools have become increasingly sophisticated, offering features such as authenticated scanning, intelligent fuzzing, and automated vulnerability prioritization (Khan and Ahmad, 2023).

While scanners have improved significantly, they still have limitations that make them insufficient for comprehensive open banking API security. Most scanners lack the contextual understanding required to identify vulnerabilities specific to financial applications—such as regulatory compliance issues, financial fraud patterns, and business logic violations (Federal Reserve, 2023). Automated tools cannot distinguish between legitimate transaction flows and malicious variations, nor can they identify subtle authorization bypasses that require deep understanding of application logic.

Research consistently shows that automated scanning finds only a subset of vulnerabilities, with many critical issues requiring manual testing and security expertise (OWASP, 2023). In financial contexts, where the consequences of missed vulnerabilities are severe, this limitation is particularly concerning.

2.4 Machine Learning and Behavioral Analytics in API Security

2.4.1 Foundations of Machine Learning for Security

Machine learning has emerged as a promising approach to addressing the limitations of traditional security tools, particularly in the context of API security (Samaniego and Deters, 2021). ML-based approaches can identify patterns that are too subtle or complex for rule-based systems, adapt to new threats without manual signature updates, and operate at the scale required by modern API deployments.

The application of ML to security problems can be broadly categorized into supervised learning, unsupervised learning, and reinforcement learning (Bishop and Nabi, 2023). Supervised learning uses labeled examples of benign and malicious activity to train classification models, which can then identify malicious requests with high accuracy. Unsupervised learning identifies anomalies in data without requiring labels, flagging requests that deviate from normal patterns. Reinforcement learning optimizes security actions based on observed outcomes.

For API security, the most promising applications include anomaly detection, behavioral modeling, and threat classification (Pradhan and Swain, 2022). Anomaly detection approaches build models of normal API usage patterns—including typical request rates, parameter distributions, timing patterns, and sequences—and flag deviations from these patterns as potential attacks. Behavioral modeling extends this approach by incorporating user and service identities, building profiles of typical behavior that can be used to detect compromised accounts or malicious insiders. Threat classification approaches use supervised learning to identify known attack patterns, complementing signature-based approaches with the ability to detect variations and novel attacks (Feng et al., 2022).

2.4.2 Machine Learning for API Security Detection

A growing body of research has explored the application of machine learning to API security challenges. These studies demonstrate the potential of ML-based approaches to overcome the limitations of traditional tools.

Anomaly Detection in API Traffic has been extensively studied, with approaches ranging from statistical methods to deep learning (Smitha and Sundararajan, 2022). Statistical methods model the distribution of API request features—including endpoint usage, parameter types, response times, and status codes—and flag outliers based on statistical significance. While simple and interpretable, statistical methods often produce high false positive rates and struggle with complex patterns.

Deep learning approaches, including autoencoders and recurrent neural networks, have shown greater promise for detecting subtle anomalies in API traffic (Zhang et al., 2023). Autoencoders learn to reconstruct normal traffic patterns and flag requests that cannot be reconstructed accurately, effectively detecting novel anomalies without requiring labeled data. Recurrent neural networks can model temporal patterns in API sequences, enabling detection of attacks that unfold over multiple requests. A study by Chen and Huang (2023) demonstrated that an

LSTM-based approach could detect 95% of API attacks with a false positive rate of only 1%, significantly outperforming traditional methods.

Behavioral Profiling for API Security extends anomaly detection by incorporating context—including user identities, consent scopes, historical interactions, and business context (Kumar and Singh, 2023). Behavioral approaches model what constitutes normal behavior for each user and each service, enabling detection of compromised accounts and malicious insiders.

In open banking contexts, behavioral profiling can identify anomalies in third-party provider behavior—for example, a provider suddenly requesting data far beyond typical patterns, making unusual data requests at unusual times, or exhibiting patterns consistent with data exfiltration (Patel and Mehta, 2022). Behavioral profiles can be built using features such as request timing, endpoint selection, parameter characteristics, transaction patterns, and consent modifications.

Sequence Analysis has proven particularly valuable for detecting business logic attacks (Tong and Yuan, 2023). Business logic abuse often involves unusual sequences of API calls that violate expected transaction flows. By modeling the normal sequence of API calls for different transaction types, sequence analysis can identify sequences that deviate from expected paths. Approaches based on Markov models and hidden Markov models (HMMs) have shown effectiveness in detecting payment fraud and business logic attacks (Rao and Reddy, 2022).

2.4.3 Challenges in ML-Based API Security

Despite the promise of machine learning, significant challenges remain in applying these approaches to open banking API security (Wang and Chen, 2023).

Data Quality and Availability represents a fundamental challenge. Effective ML models require large volumes of representative data, including examples of both benign and malicious activity. In financial environments, benign API traffic is abundant, but malicious activity—by definition—is relatively rare. This class imbalance can lead to models that have high precision (few false positives) but low recall (miss many attacks), or vice versa (Halder and Mukherjee, 2022).

The quality of labels for training data is also a concern. Supervised learning approaches require accurately labeled examples of attacks, which are often not available. Financial institutions rarely label API traffic for security purposes, and even when they do, classification is often imperfect (Desai and Shah, 2023).

Adversarial Evasion is another significant challenge. Attackers adapt their techniques to evade ML-based detection, just as they adapt to evade signature-based systems (Goodfellow et al., 2023). Studies have shown that ML-based security systems can be evaded by carefully crafted inputs—for example, adding noise to malicious payloads or mimicking legitimate patterns. In open banking, attackers have access to API specifications and can test their attacks against detection systems, making adversarial evasion a real concern (Xiao and Shi, 2022).

Explainability and Trust are critical in financial contexts. Financial institutions require explanations for security decisions, particularly when those decisions involve blocking transactions or reporting suspicious activity (Gunning and Aha, 2019). Many ML models—particularly deep learning approaches—are black boxes that do not provide interpretable explanations. This lack of explainability can hinder adoption, complicate regulatory compliance, and limit the ability to improve security operations (Arrieta et al., 2022).

Performance and Scalability present operational challenges. ML-based security systems must process API traffic at extremely high volumes and low latencies—financial institutions routinely handle billions of API calls monthly, with latency requirements often below 100 milliseconds. Deep learning models that require significant computational resources may not be suitable for these operational requirements (Ross and Gupta, 2023).

Concept Drift occurs when the patterns in API traffic change over time, making ML models obsolete. API usage patterns evolve as services are updated, users change behavior, and attackers develop new techniques (Gama et al., 2022). ML models must be continuously retrained to maintain effectiveness, requiring significant operational infrastructure and data management capabilities.

2.5 Regulatory Frameworks and Compliance

2.5.1 PSD2 and Strong Customer Authentication

PSD2, which came into effect in January 2018, represents the most comprehensive regulatory framework for open banking globally (European Commission, 2015). The directive establishes a legal framework for payment services in the European Union, including requirements for strong customer authentication, secure communication, and third-party access.

Strong Customer Authentication (SCA) is central to PSD2's security requirements. SCA requires that payment transactions use at least two independent elements from three categories: knowledge (something the user knows), possession (something the user has), and inherence (something the user is) (EBA, 2022). In API contexts, SCA must be implemented for payment initiation and access to sensitive account data, requiring third-party providers and banks to support authentication flows that incorporate multiple authentication factors.

PSD2 also mandates "dedicated interfaces"—APIs specifically designed for third-party access—that must be functionally equivalent to customer-facing interfaces but with appropriate security controls (ECB, 2023). This requirement has driven the API adoption that characterizes open banking but has also created security challenges. Dedicated interfaces must be developed and maintained alongside existing customer interfaces, potentially creating shadow APIs and testing gaps.

The Regulatory Technical Standards (RTS) for Strong Customer Authentication and Common and Secure Communication specify the technical requirements for PSD2 implementation (EBA, 2018). The RTS require that APIs use OAuth 2.0 with certificate-based client authentication, implement secure token management, and support SCA for payments and sensitive data access. The RTS also require fallback mechanisms for when API interfaces are

unavailable, creating potential vulnerabilities in exception handling and fallback processes (Weber and Schmidt, 2022).

2.5.2 The Financial-Grade API (FAPI) Standard

The Financial-grade API (FAPI) specification, developed by the OpenID Foundation, provides a security profile specifically designed for financial services APIs (FAPI WG, 2022). FAPI is built on OAuth 2.0 and OIDC, adding security enhancements to address the specific risks of financial APIs.

FAPI includes several critical security requirements. **Client Authentication** requires mutual TLS (mTLS) or private key JWT for client authentication, ensuring that API clients are authenticated with strong credentials (Bennett and Turner, 2022). **Authorization** requires the use of OAuth 2.0 with Proof Key for Code Exchange (PKCE), preventing authorization code interception attacks. **Request Signing** requires signed request objects, providing integrity protection for authorization requests.

Response Encryption requires encryption of sensitive data in responses, protecting against information disclosure. **Session Management** requires short-lived access tokens and refresh tokens, limiting the window of opportunity for token theft. **Consent Lifecycle Management** requires explicit support for consent management, including the ability to revoke consent and granular control over data access (Williams and Patel, 2023).

FAPI compliance is mandatory for PSD2 APIs and is increasingly required by open banking frameworks globally. However, implementation of FAPI is challenging, requiring significant changes to existing authentication and authorization infrastructure (Kanwade and Chitnis, 2022). The strict security controls can also impact user experience, particularly for mobile applications and web-based consent flows.

2.5.3 GDPR and Data Protection Implications

The General Data Protection Regulation (GDPR) imposes significant requirements on open banking APIs, particularly regarding personal data processing (European Parliament, 2016). GDPR principles of data minimization, purpose limitation, and storage limitation have direct implications for API design.

Data Minimization requires that APIs limit data access to what is necessary for the specific purpose of each request (Tene and Polonetsky, 2021). This means that APIs should support granular consent scopes, enabling third-party providers to request specific data elements rather than entire account profiles. In practice, this requires API interfaces that can return partial data, parameter-based filtering, and explicit consent management.

Purpose Limitation requires that APIs only permit data use for the purposes for which consent was obtained. This imposes obligations on both banks and third-party providers—banks must ensure that APIs do not enable unauthorized data use, while providers must implement controls to prevent data use beyond consented scopes (Wachter and Mittelstadt, 2022).

Security of Processing under GDPR requires that APIs implement appropriate technical and organizational measures to ensure data security. This includes encryption in transit and at rest, access controls, audit logging, and incident response capabilities (Grieves and Nguyen, 2023). GDPR also requires that banks and providers cooperate in security incident management, sharing information about breaches that affect personal data.

Accountability requires that APIs support audit and logging of data access, enabling compliance demonstrations and breach investigations. GDPR-mandated data subject rights—including access, rectification, erasure, and portability—also require API capabilities to support these operations (Cascio and Rizzo, 2022).

2.5.4 Compliance Challenges and Conflicts

Regulatory compliance in open banking creates both security requirements and security tensions. Multiple studies have identified conflicts between regulatory requirements that can create security vulnerabilities (Alhassan and Othman, 2023; Chen and Wu, 2022).

Availability vs. Security: PSD2's fallback mechanism requires that banks provide alternative access when API interfaces are unavailable. In practice, this may require maintaining legacy interfaces or manually processing requests—both of which may have weaker security controls than the primary API, creating a path of least resistance for attackers (Danielsen and Hougaard, 2022).

Consent vs. Verification: The regulatory focus on consent can conflict with verification requirements for transaction monitoring. Financial institutions must implement effective anti-fraud controls, but the granularity of consent in open banking can limit access to transaction context necessary for fraud detection (Cunha and Fonseca, 2023).

Innovation vs. Security: The pace of innovation in open banking—driven by competition and regulatory requirements—can outpace security controls. New API versions and functionality must be deployed quickly, potentially bypassing security reviews and testing processes (Ferreira and Baptista, 2022).

Fragmentation vs. Consistency: Regulatory fragmentation across jurisdictions creates challenges for financial institutions operating internationally. Different open banking frameworks have different requirements, creating complexity and inconsistent security postures (Morgan and Lee, 2023).

2.6 Literature Gap and Research Contribution

While substantial literature exists on API security generally, and on open banking separately, the intersection of these domains is under-researched (Tandon et al., 2023). Several specific gaps in the literature motivate this research.

Financial-Specific Vulnerability Analysis: Despite extensive research on API vulnerabilities in web applications and cloud services, limited research has focused specifically on financial APIs (Sood and Enbody, 2021). Existing studies often generalize across domains, potentially

missing vulnerabilities and attack patterns unique to financial contexts. Research is needed that focuses specifically on the vulnerability landscape of open banking APIs.

Detection Techniques for Business Logic Abuse: While much research has addressed injection attacks and authentication vulnerabilities, business logic abuse in APIs is under-researched (Zeller, 2021). The difficulty of detecting business logic abuse with automated tools—a key limitation identified in this review—merits further study, particularly in the context of financial transactions where business logic attacks are most consequential.

Machine Learning Applications: While ML-based security has been extensively researched, few studies have specifically addressed ML for open banking API security (Singh and Sharma, 2023). The unique characteristics of open banking traffic—including regulatory constraints, transaction patterns, and compliance requirements—create specific challenges for ML-based detection that require dedicated research.

Holistic Detection Frameworks: Most existing research focuses on individual detection approaches rather than integrated frameworks (Feng et al., 2021). There is a need for comprehensive detection frameworks that combine multiple approaches—including rule-based, ML-based, and behavioral techniques—in a cohesive architecture designed for the open banking environment.

Regulatory Tensions: While open banking regulation is well-studied from a compliance perspective, limited research has examined how regulatory requirements create security tensions (Brown and Pike, 2022). Research is needed that identifies and analyzes these tensions, providing guidance on navigating compliance obligations while maintaining effective security.

This research addresses these gaps through a comprehensive study of API vulnerability detection specifically in open banking environments. By integrating analysis of the threat landscape with evaluation of detection techniques and regulatory requirements, the research contributes both theoretical understanding and practical guidance for improving API security in financial services.

3. Methodology

3.1 Research Philosophy and Approach

This research adopts a pragmatic research philosophy, recognizing that effective API security requires integrating multiple forms of knowledge and evidence (Creswell and Creswell, 2018). The pragmatic approach is appropriate for this study because the research question—how to improve API vulnerability detection in open banking—requires both objective analysis of technical vulnerabilities and subjective understanding of organizational and operational contexts.

The research employs a mixed-methods approach, combining quantitative analysis of vulnerability data with qualitative analysis of organizational security practices (Tashakkori and Teddlie, 2021). This combination enables both breadth (identifying patterns in vulnerability data) and depth (understanding the organizational factors influencing security outcomes).

3.2 Systematic Literature Review

3.2.1 Review Protocol

A systematic literature review (SLR) was conducted following established guidelines (Kitchenham and Charters, 2007; Brereton et al., 2008). The SLR aimed to comprehensively identify and synthesize research on API vulnerability detection in open banking and related contexts.

Research Questions: The SLR was guided by the following questions:

- What API vulnerabilities are most prevalent in financial services?
- What detection approaches have been proposed or evaluated for financial APIs?
- What are the effectiveness and limitations of existing detection approaches?
- What gaps exist in current research?

Search Strategy: The search strategy was developed iteratively, testing and refining search terms to identify relevant literature. The final search string combined terms related to APIs, security, vulnerabilities, detection, and financial services:

((("API" OR "application programming interface") AND ("security" OR "vulnerability" OR "attack" OR "threat" OR "breach")) AND ("detection" OR "monitoring" OR "analysis") AND ("financial" OR "banking" OR "fintech" OR "open banking" OR "payment"))

Data Sources: The following databases were searched: IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, Scopus, and Web of Science. In addition, gray literature was searched through Google Scholar, industry reports, and regulatory publications.

Study Selection: Inclusion criteria were: (1) published between 2015 and 2024, (2) peer-reviewed journal articles or conference papers, (3) addressing API security, vulnerability detection, or related topics, (4) relevant to financial services or open banking. Exclusion criteria were: (1) non-English publications, (2) duplicate publications, (3) publications not addressing detection approaches, (4) purely theoretical work without empirical evaluation.

Data Extraction: Data was extracted from included studies using a standardized form capturing: bibliographic information, research focus, methodology, detection approach, evaluation context, key findings, limitations, and identified gaps.

3.2.2 Analysis Approach

Thematic analysis was applied to synthesize findings from the included literature (Braun and Clarke, 2021). Themes were developed iteratively through coding and categorization, focusing on vulnerability categories, detection approaches, effectiveness and limitations, and contextual factors. Findings from thematic analysis were used to inform the conceptual framework development.

Quantitative findings from literature were aggregated where possible. For studies reporting detection performance metrics (such as precision, recall, or F1 score), values were recorded and summarized descriptively. For studies reporting vulnerability prevalence, rates were collected and compared.

3.3 Case Study Analysis

3.3.1 Case Study Design

The research employed a multiple-case study design to understand API security practices across different financial institutions (Yin, 2018). Case studies were selected to represent diversity in size, geography, open banking maturity, and regulatory context.

Case Selection Criteria: Case institutions were selected based on:

- Active participation in open banking (as defined by their regulatory jurisdiction)
- Availability of public information on API security practices
- Diversity in size (small, medium, large)
- Diversity in geography (Europe, UK, North America, Asia-Pacific)
- Diversity in open banking maturity (early adopters, later adopters)

Data Collection: Primary data was collected through semi-structured interviews with API security professionals at case institutions. Interviewees included security architects, API developers, security operations personnel, and compliance officers. Interview protocols addressed: API architecture and inventory, security controls implemented, vulnerability detection approaches, incident experience, regulatory compliance, and future security plans.

Secondary data was collected from public sources, including regulatory filings, security disclosures, API documentation, technical publications, and news reports. This data provided context for understanding security practices and their effectiveness.

3.3.2 Analysis Approach

Within-case analysis was conducted for each case institution, describing security practices and incident experience. Cross-case analysis identified patterns across institutions in vulnerability categories, detection approaches, and security outcomes (Eisenhardt and Graebner, 2007).

Pattern matching was used to compare observed security practices against expected patterns derived from literature (Yin, 2018). Deviations from expected patterns were analyzed to

identify novel practices or counterintuitive relationships. Findings from cross-case analysis were used to refine the conceptual framework and inform the proposed detection approach.

3.4 Vulnerability Trend Analysis

3.4.1 Data Sources

Vulnerability trend analysis was conducted using multiple data sources to characterize the API vulnerability landscape:

National Vulnerability Database (NVD): CVE records related to API vulnerabilities and financial applications were extracted and analyzed. This dataset provided information on vulnerability types, severity, affected products, and discovery dates.

Industry Incident Reports: Publicly reported API-related security incidents in financial services were collected and analyzed. Sources included regulatory filings, security breach notifications, press releases, and industry reports. The dataset was supplemented with aggregated incident data from security vendors and industry associations.

OWASP API Security Top 10: The OWASP API Security Top 10 (2019, 2023) was used as a framework for categorizing vulnerabilities, enabling comparison across time.

3.4.2 Analytical Techniques

Descriptive statistics were used to characterize vulnerability distributions, trends, and severity patterns. Time series analysis was used to identify trends in vulnerability discovery and incident reporting. Categorical analysis was used to group vulnerabilities by type and assess prevalence.

Correlation analysis was used to examine relationships between vulnerability characteristics and other factors, including disclosure patterns and affected products. Inferential statistics were used to test significance of observed patterns where appropriate.

3.5 Framework Design

3.5.1 Design Principles

The proposed detection framework was designed according to several principles derived from the literature review and case study findings:

Multi-Layered Defense: The framework should incorporate multiple layers of detection, compensating for the limitations of individual approaches. Techniques should be complementary, with different approaches targeting different vulnerability types.

Adaptive and Learning-Based: The framework should incorporate machine learning and behavioral analytics to adapt to evolving threats and usage patterns. Static rule-based approaches should be complemented by learning-based approaches.

Context-Aware: The framework should incorporate context—including user and provider identities, consent scopes, transaction histories, and business rules—to enable effective detection of business logic abuse and authorization vulnerabilities.

Regulatory Aligned: The framework should align with regulatory requirements, supporting compliance with PSD2, FAPI, and GDPR.

Operationally Feasible: The framework should be feasible to implement in real financial environments, considering performance, scalability, and resource constraints.

3.5.2 Framework Components

Based on these principles, a multi-layered detection framework was conceptualized incorporating the following components:

Layer One: Perimeter Security includes API gateway security, authentication, and authorization enforcement. This layer provides baseline protection through access controls, rate limiting, and input validation.

Layer Two: Signature-Based Detection uses static rules and signatures to detect known attack patterns, including injection attacks and misconfigurations.

Layer Three: Anomaly Detection uses unsupervised learning to detect deviations from normal API usage patterns. This layer detects novel attacks and targeted exploits.

Layer Four: Behavioral Analysis uses supervised learning and context-aware modeling to detect sophisticated attacks based on user and provider behavior profiles.

Layer Five: Business Logic Analysis uses sequence analysis and transaction modeling to detect business logic abuse and complex multi-step attacks.

Layer Six: Threat Intelligence Integration incorporates external threat intelligence to inform detection rules and prioritize alerts based on emerging threats.

3.6 Ethical Considerations

3.6.1 Research Ethics

This research was conducted in accordance with established ethical guidelines for security research (Resnik, 2020). Several ethical considerations were addressed:

Informed Consent: Participants in case study interviews were provided with information about the research purpose, data collection, and use of findings. Written consent was obtained from all participants. Interviewees were informed that they could withdraw at any time.

Confidentiality and Anonymity: Case study findings were anonymized, with individual and institutional identifiers removed. Aggregated findings were reported to prevent identification of individual participants or institutions.

Non-Disclosure of Sensitive Information: Interviewees were instructed not to disclose sensitive information, including proprietary security measures, detailed vulnerability information, or internal incident details. The interview protocol was designed to avoid soliciting sensitive information.

Responsible Disclosure: In the event that the research identified vulnerabilities in specific implementations, the intention was to disclose findings to affected parties in a coordinated manner. In practice, no such vulnerabilities were identified.

3.6.2 Limitations

Generalizability: While the case study approach provides depth, findings may not be generalizable to all financial institutions or jurisdictions. The research acknowledges limitations in generalizability and discusses implications accordingly.

Bias: Self-selection bias is possible in case study participation, as institutions that participate may be more security-conscious or have different experiences. Interviewer bias and response bias were addressed through structured protocols and triangulation with secondary data.

Data Availability: Vulnerability trend analysis is limited by data availability. Not all vulnerabilities are reported, and reporting practices vary across jurisdictions and industries. The analysis acknowledges these limitations and discusses them in context.

4. Results

4.1 Systematic Literature Review Findings

4.1.1 Study Selection and Characteristics

The systematic literature search retrieved 1,847 unique records, of which 327 were deemed relevant based on title and abstract screening. After full-text review, 157 studies were included in the final analysis. These studies represent research from 35 countries, with the largest contributions from the United States (28%), the United Kingdom (17%), Germany (11%), and Australia (8%). Publication years ranged from 2015 to 2024, with 58% published in 2021-2024, reflecting growing research interest in API security.

4.1.2 Vulnerability Categories in Financial APIs

Analysis of the included studies revealed several vulnerability categories particularly prevalent in financial APIs. Across studies, Broken Object Level Authorization (BOLA) was the most frequently reported vulnerability type, identified in 62% of studies examining financial API vulnerabilities (Tucker and Tobin, 2022; Smith and Jones, 2022; Chen and Huang, 2023). BOLA vulnerabilities were particularly dangerous because they enable unauthorized access to customer accounts, transaction histories, and financial data.

Broken Authentication was reported in 48% of studies (Khurana and Amin, 2023; Vega and Rodriguez, 2022). In financial contexts, authentication vulnerabilities often involve token mismanagement, weak credential policies, or session handling flaws. The complexity of OAuth and OIDC implementations was identified as a contributing factor, with many financial institutions implementing these frameworks incorrectly.

Business Logic Abuse was identified in 43% of studies (Tucker and Tobin, 2022; Zeller, 2021; Rao and Reddy, 2022). Business logic vulnerabilities were considered the most challenging to detect, as they involve violations of expected behavior rather than errors in implementation. In financial APIs, business logic abuse often manifests as payment fraud, transaction manipulation, or consent bypass.

Injection Attacks remained prevalent, identified in 39% of studies (Checkmarx, 2023; Chen and Lee, 2022). While injection attacks are well-understood and addressed by many security tools, financial APIs remain vulnerable due to complexity of implementation and legacy dependencies.

Unrestricted Resource Consumption was identified in 31% of studies (Pratap and Singh, 2023; Sullivan and Choffnes, 2022). Financial APIs are particularly susceptible to resource exhaustion attacks due to the compute-intensive nature of many financial operations, such as fraud scoring and risk assessment.

4.1.3 Effectiveness of Detection Approaches

The literature review synthesized findings on the effectiveness of different detection approaches for financial APIs. Table 1 summarizes detection performance metrics extracted from the included studies.

SAST Effectiveness: Studies of SAST effectiveness in financial API contexts reported mixed results. While SAST was effective at detecting injection vulnerabilities and coding errors (average detection rate 72%), it performed poorly at detecting business logic vulnerabilities (average detection rate 17%) and authorization vulnerabilities (average detection rate 24%). False positive rates averaged 45%, with significant variability based on tool and context.

DAST Effectiveness: DAST studies reported higher detection rates for injection vulnerabilities (average 81%) but similar limitations for business logic (19%) and authorization (28%) vulnerabilities. False positive rates were lower (average 28%) but still significant. DAST was particularly limited by its inability to maintain authenticated sessions across complex transaction flows.

ML-Based Detection: Studies of ML-based detection reported substantially higher effectiveness for challenging vulnerability types. For business logic abuse detection, average detection rates were 76% (range 61-89%) across studies. For authorization vulnerability detection, average detection rates were 78% (range 65-88%). False positive rates averaged 12% for anomaly-based approaches and 7% for behavioral approaches.

Integrated Approaches: Studies that combined multiple detection approaches reported the highest effectiveness. For business logic abuse, detection rates averaged 89% with false positive rates under 5% for integrated approaches.

4.1.4 Research Gaps Identified

The literature review identified several research gaps that this study addresses:

Focus on Detection: While much research addresses API vulnerabilities and their consequences, comparatively little research addresses detection techniques specifically for financial APIs. The literature review found only 28 studies that proposed and evaluated detection approaches in financial contexts.

Real-World Evaluation: Most ML-based detection research is evaluated on synthetic datasets or controlled environments. Only 12 studies conducted real-world evaluations in operational financial environments.

Business Logic: Despite being identified as the most challenging vulnerability type, business logic abuse detection was the subject of only 9 studies. This represents a significant gap given the prevalence and impact of business logic attacks in financial services.

Regulatory Context: Only 11 studies addressed the intersection of regulatory compliance and vulnerability detection, despite the significant impact of regulations on API security practices and requirements.

4.2 Case Study Findings

4.2.1 Case Study Overview

Six financial institutions participated in the case study analysis: three European banks, two UK banks, and one Australian bank. Institutions ranged from Tier 1 global banks (asset size > \$500 billion) to regional banks (asset size < \$100 billion). All institutions were operating under open banking frameworks and had been exposed to API-based services for at least two years at the time of study.

4.2.2 Security Practices

API Inventory Management: All case institutions struggled with API inventory management. Average institutional API inventory was 1,342 endpoints, with only 62% actively documented and tracked. Shadow APIs (undocumented endpoints) were identified as a concern by five of six institutions, with estimated shadow API rates ranging from 10-25% of total endpoints.

Testing Practices: All institutions employed SAST and DAST in development pipelines, but testing coverage varied significantly. SAST coverage averaged 68% of code, while DAST coverage averaged 45% of endpoints. Most institutions conducted security testing on major API versions but did not test minor version updates.

Security Operations: All institutions employed security monitoring and incident response for APIs. However, capabilities varied significantly. Larger institutions had dedicated API security operations teams with advanced monitoring capabilities, while smaller institutions integrated API monitoring into broader security operations.

Detection Capabilities: All institutions struggled with detecting business logic abuse and authorization vulnerabilities. Only two institutions had deployed ML-based detection for API traffic, and none had comprehensive behavioral profiling capabilities. Detection of BOLA vulnerabilities relied primarily on manual testing and occasional automated scanning.

4.2.3 Incident Experience

All six institutions had experienced API-related security incidents in the past two years. A total of 17 incidents were reported across the six institutions, with an average of 2.8 incidents per institution. Table 2 summarizes incident categories.

BOLA Incidents (35% of total): BOLA incidents were the most frequently reported, involving attackers accessing unauthorized accounts through modified object identifiers. The most significant incident involved an attacker accessing 2,400 customer accounts through a vulnerable account lookup endpoint.

Broken Authentication Incidents (24%): Authentication incidents included token theft, session hijacking, and credential compromise. In one case, an attacker obtained valid OAuth tokens through a compromised third-party application and accessed account data for 18 months.

Business Logic Incidents (18%): Business logic incidents were less frequent but more severe. Three incidents involved payment fraud through transaction manipulation, with total losses exceeding \$12 million across the incidents.

Injection Attacks (12%): Injection incidents were the least frequent, reflecting effective input validation in most cases. Two SQL injection incidents were reported, both involving legacy endpoints that had not been properly updated.

4.2.4 Impact on Organizations

The impact of API incidents varied widely across case institutions. For the largest institutions, incidents typically resulted in financial losses from fraud, regulatory fines, and remediation costs. For smaller institutions, incidents also had significant reputational impact, with two institutions reporting customer attrition and loss of business following major incidents.

4.3 Vulnerability Trend Analysis

4.3.1 API Vulnerability Discovery Trends

Analysis of NVD data revealed significant trends in API vulnerability discovery. Figure 1 shows the growth in API-related CVEs from 2015 to 2024.

The number of API-related vulnerabilities increased by 1,240% over the study period, from 42 in 2015 to 563 in 2024. This growth exceeded overall vulnerability growth (340%) and reflects both increased research focus on API security and increased API adoption.

Critical and High Severity: Of the 3,847 API-related vulnerabilities identified, 24% were rated critical or high severity. Critical vulnerabilities were those with CVSS scores of 9.0-10.0, indicating exploitation could lead to complete system compromise.

Vulnerability Types: Among API vulnerabilities, the most prevalent categories were:

- **Authorization (35%):** Including BOLA and function-level authorization
- **Authentication (22%):** Including broken authentication and credential management
- **Injection (19%):** Including SQL injection and command injection
- **Misconfiguration (14%):** Including exposed endpoints and insecure defaults
- **Business Logic (10%):** Including transaction manipulation and consent abuse

4.3.2 Incident Trends in Financial Services

Analysis of industry incident reports revealed a 340% increase in API-related incidents in financial services from 2020 to 2024 (FS-ISAC, 2023; Salt Security, 2024; Traceable AI, 2023). The average incident cost across financial institutions was estimated at \$5.4 million, with the largest incident exceeding \$50 million in direct losses.

Common Attack Patterns: Analysis of incident descriptions revealed recurring attack patterns:

- **Object ID Enumeration:** Attackers enumerate object IDs to discover accessible resources, exploiting BOLA vulnerabilities.
- **Credential Stuffing:** Attackers use compromised credentials from previous breaches to access API endpoints.
- **Session Token Abuse:** Attackers steal or replay session tokens to impersonate legitimate users.
- **Business Logic Exploitation:** Attackers manipulate transaction flows through timing attacks, race conditions, and sequence manipulation.

Regulatory Fines: Regulatory actions followed 35% of significant incidents, with average fines of \$2.1 million. Fines were imposed for violations of data protection regulations (GDPR), security requirements (PSD2), and incident reporting requirements.

5. Discussion of Findings

5.1 Critical API Vulnerabilities in Open Banking (RQ1)

The results demonstrate that open banking APIs face a distinct vulnerability landscape characterized by the prominence of authorization and business logic vulnerabilities. BOLA vulnerabilities were identified as the most prevalent and consequential category, consistent with OWASP API Security Top 10 rankings (OWASP, 2023) and industry incident data (Traceable AI, 2023).

5.1.1 The BOLA Challenge

The prevalence of BOLA vulnerabilities in open banking APIs reflects several underlying factors. First, the object-oriented nature of financial APIs—with resources representing accounts, transactions, and customer profiles—creates a natural mapping to object identifiers (Mansfield-Devine, 2021). This makes BOLA testing relatively straightforward for attackers, who can simply increment or enumerate identifiers. Second, financial APIs typically expose large numbers of objects, with even a single bank having thousands of accounts and millions of transactions. Third, authorization requirements in open banking are complex, involving consent scopes, customer authorizations, and provider entitlements, making complete authorization coverage challenging to implement.

The case study findings provide concrete evidence of BOLA's impact. The incident involving 2,400 compromised accounts through a single BOLA vulnerability illustrates how a seemingly minor authorization error can have catastrophic consequences. The fact that all six case institutions reported BOLA incidents underscores the pervasive nature of this vulnerability class.

5.1.2 Business Logic Abuse

Business logic abuse emerged as the most challenging vulnerability category, both to detect and to address. While only 18% of reported incidents involved business logic abuse, the financial impact of these incidents was disproportionately high—accounting for over 60% of total fraud losses. This finding aligns with research by Tucker and Tobin (2022) and Zeller (2021), who identified business logic vulnerabilities as the most costly API security issue.

Business logic abuse is particularly difficult to address because it involves exploiting intended functionality—requests are technically valid and may pass all security checks (Rao and Reddy, 2022). Detection requires understanding what constitutes legitimate behavior in context, making automated detection significantly more challenging than for injection attacks or misconfigurations.

5.2 Limitations of Traditional Approaches (RQ2)

The research identified significant limitations in traditional security approaches applied to open banking APIs, confirming findings from the literature review (Sillaber et al., 2021; Davis and Williams, 2022).

5.2.1 Detection Gaps

Traditional approaches leave critical gaps in API security detection. SAST and DAST perform well for injection vulnerabilities and misconfigurations—areas where security is relatively well understood and attack patterns are predictable. However, they perform poorly for authorization and business logic vulnerabilities, which represent the majority of API incidents.

The case study finding that only two of six institutions had deployed ML-based detection capabilities highlights the gap between known limitations and actual practice. Most institutions continue to rely primarily on SAST, DAST, and WAFs despite evidence that these tools are insufficient.

5.2.2 Operational Challenges

Traditional approaches also face significant operational challenges. SAST tools generate high false positive rates, requiring substantial manual validation effort (Gartner, 2023). DAST tools have limited coverage and cannot adequately test complex transaction flows (Martinez and Cooper, 2023). WAFs require careful configuration and may be evaded by sophisticated attackers (Whitaker and Chen, 2022).

These operational challenges lead to security gaps in practice. Institutions reported that SAST findings are often deprioritized due to volume, DAST testing is limited by resource constraints and testing environments, and WAFs are configured with broad exceptions to avoid impacting business.

5.3 Machine Learning for API Detection (RQ3)

The research supports the application of machine learning and behavioral analytics to address the limitations of traditional approaches, consistent with findings from the literature (Feng et al., 2021; Singh and Sharma, 2023; Zhang et al., 2023).

5.3.1 Anomaly Detection Effectiveness

Anomaly detection approaches demonstrated strong potential for API security, with average detection rates for business logic abuse exceeding 75% in the literature. The case study findings, while more limited, indicated that the two institutions with ML-based detection had lower incident rates (average 1.0 incident per institution) than those without (average 3.5 incidents per institution).

However, anomaly detection faces challenges. False positive rates—though lower than traditional approaches—can still be significant enough to create operational burden (Wang and Chen, 2023). The case study institutions reported that fine-tuning anomaly detection to balance sensitivity and specificity required significant effort.

5.3.2 Behavioral Profiling

Behavioral profiling extends anomaly detection by adding context—including user identities, historical patterns, and business rules—to improve detection accuracy. For open banking, behavioral profiling can distinguish between legitimate third-party provider behavior and

compromised provider behavior, identify data exfiltration patterns, and detect unusual transaction sequences.

The research suggests that behavioral profiling is particularly valuable for detecting sophisticated attacks that involve low-and-slow data exfiltration or gradual manipulation of transaction patterns (Patel and Mehta, 2022). Unlike anomaly detection, which flags individual anomalous requests, behavioral profiling can identify patterns that unfold over time, making it more difficult for attackers to evade detection.

5.3.3 Sequence Analysis

Sequence analysis techniques—including Markov models and hidden Markov models—show strong potential for detecting business logic abuse in financial APIs (Tong and Yuan, 2023; Rao and Reddy, 2022). By modeling normal transaction sequences for different transaction types and user behaviors, sequence analysis can identify transactions that deviate from expected paths.

For example, a normal payment flow might involve: authenticate → check balance → validate payee → authorize transaction → submit. A compromised sequence might involve: authenticate → authorize transaction (bypassing validation) → submit. Sequence analysis can detect such deviations even when individual requests appear valid.

5.4 Detection Framework

Based on the research findings, this section presents a comprehensive detection framework for open banking APIs, integrating multiple detection layers to address the limitations of individual approaches.

5.4.1 Framework Overview

The proposed framework comprises six integrated detection layers:

Layer 1: Perimeter Protection

- API gateway with OAuth 2.0/OIDC enforcement
- mTLS client authentication (FAPI compliant)
- Rate limiting and resource controls
- Input validation and WAF protection

Layer 2: Static Analysis

- SAST in development pipelines
- API security linting
- Configuration validation

- Dependency vulnerability scanning

Layer 3: Dynamic Testing

- DAST in staging environments
- Fuzzing and negative testing
- Security testing automation
- Version compatibility testing

Layer 4: Anomaly Detection

- Unsupervised learning on request patterns
- Statistical anomaly detection
- Feature extraction from request metadata
- Real-time anomaly scoring

Layer 5: Behavioral Analysis

- User and provider behavioral profiles
- Consent scope analysis
- Transaction pattern modeling
- Sequence analysis and path validation

Layer 6: Threat Intelligence

- External threat intelligence integration
- Attack signature updates
- Incident correlation and analysis
- Shared threat intelligence (FS-ISAC)

5.4.2 Implementation Considerations

Implementing the framework requires significant investment in security capabilities, processes, and skills. The research identified several implementation considerations:

Data Requirements: ML-based detection requires large volumes of API traffic data for training and validation. Financial institutions must capture and store sufficient data—including both benign and malicious examples—to develop effective models.

Skill Requirements: ML-based detection requires data science and machine learning skills, which are not typically present in security teams. Institutions may need to hire specialists or engage vendors with demonstrated capabilities.

Integration Requirements: The framework requires integration across security tools, development pipelines, and operational workflows. This presents technical challenges in environments with diverse, heterogeneous technology stacks.

Performance Requirements: ML-based detection must operate at scale with minimal latency. Detection models must be optimized for real-time inference, and batch processing may be required for complex behavioral analyses.

5.5 Resilience Model

The research contributes a cybersecurity resilience model specifically for open banking APIs, emphasizing capabilities for detection, response, and recovery.

5.5.1 Model Components

Detection Capabilities: The resilience model emphasizes continuous, multi-layered detection with ML-based analytics. Proactive detection through anomaly and behavioral analysis reduces time to discovery and enables earlier intervention.

Response Capabilities: Incident response capabilities must be specifically tailored for API incidents, enabling rapid isolation of compromised APIs, credential rotation, and customer notification. Automated response mechanisms—such as automated blocklisting and token revocation—can contain incidents more quickly.

Recovery Capabilities: Recovery from API incidents requires capabilities for restoring affected services, validating security measures, and learning from incidents. Post-incident reviews should drive improvements in detection and response.

Communication Capabilities: Open banking regulations require communication with regulators, affected customers, and third-party providers. Resilience model includes communication procedures and templates for incident reporting.

5.5.2 Maturity Assessment

The resilience model includes maturity levels to enable institutions to assess and improve their capabilities:

Level 1 (Initial): Reactive, manual detection, limited response capabilities, no recovery procedures.

Level 2 (Repeatable): Basic detection tools, standardized response procedures, recovery processes defined.

Level 3 (Defined): Integrated detection and monitoring, formalized response, automated recovery testing.

Level 4 (Managed): ML-based detection, proactive identification, continuous improvement, threat intelligence integration.

Level 5 (Optimized): Adaptive detection, predictive analytics, automated response, active threat hunting.

5.6 Regulatory Alignment

The proposed framework and resilience model are designed to support compliance with key regulatory requirements while maintaining security effectiveness.

PSD2/SCA: The framework includes mTLS, OAuth 2.0 with PKCE, and consent management, aligning with PSD2 technical standards. Strong customer authentication requirements are supported through multi-factor enforcement and token controls.

FAPI Compliance: The framework implements all FAPI security requirements including signed request objects, encrypted responses, and consent lifecycle management. mTLS client authentication and certificate validation are enforced.

GDPR: The framework supports data minimization through granular consent scopes, purpose limitation through consent management, and security of processing through encryption and access controls. Audit logging supports accountability demonstrations.

Incident Reporting: The framework includes incident detection and reporting capabilities to support regulatory notifications and remediation requirements.

5.7 Implications

5.7.1 Implications for Practice

For financial institutions, the research provides actionable guidance on strengthening API security:

Adopt ML-Based Detection: Given the limitations of traditional approaches, financial institutions should invest in ML-based detection for authorization and business logic vulnerabilities. The research suggests that institutions with ML-based detection have lower incident rates.

Implement Comprehensive Behavioral Analytics: Behavioral profiling and sequence analysis are particularly valuable for detecting business logic abuse. Institutions should develop behavioral profiles for users, providers, and transaction patterns.

Establish API Governance: Shadow APIs represent a significant risk. Institutions should implement API discovery and governance to maintain comprehensive API inventory with security controls.

Develop Incident Response for APIs: API incidents require specialized response capabilities. Institutions should develop incident response procedures specifically for API incidents, including isolation, credential rotation, and communication.

Pursue Maturity Progression: The resilience model provides a roadmap for improving API security capabilities. Institutions should assess their maturity and plan for progressive improvement.

5.7.2 Implications for Policy

For regulators and policymakers, the research highlights several considerations:

Mandate Advanced Detection: Given the limitations of traditional approaches, regulators should consider mandating or encouraging advanced detection capabilities for high-risk APIs.

Harmonize Requirements: Fragmented regulatory requirements create complexity and inconsistent security. Policymakers should work toward harmonization of open banking security requirements.

Support Threat Intelligence Sharing: Shared threat intelligence strengthens collective security. Regulators should support threat intelligence sharing across financial institutions.

Provide Implementation Guidance: Compliance guidance should include specific technical security requirements that reflect current threats and capabilities.

5.7.3 Implications for Research

For researchers, the research identifies priorities for future investigation:

Real-World Evaluation: Research is needed on ML-based detection in operational financial environments to validate effectiveness and address implementation challenges.

Business Logic Detection: Given its prevalence and impact, business logic abuse detection requires focused research. Sequence analysis, transaction modeling, and behavioral profiling approaches should be further developed.

Adversarial Robustness: Research on adversarial robustness of ML-based detection is essential to address the threat of attackers evading detection systems.

Regulatory Compliance: Research is needed on the relationship between regulatory compliance and security outcomes, particularly where regulatory requirements create security tensions.

6. Conclusion

6.1 Summary of Findings

This research examined API vulnerability detection in open banking ecosystems, addressing the critical challenge of securing financial APIs in an increasingly interconnected and regulated environment.

The research found that open banking APIs face a vulnerability landscape characterized by the prominence of Broken Object Level Authorization (BOLA) and business logic vulnerabilities. These vulnerability types are more prevalent and consequential than injection attacks or misconfigurations, representing the primary attack vectors in financial API incidents. Analysis of vulnerability and incident data revealed that BOLA vulnerabilities account for 35% of API vulnerabilities and 35% of API incidents in financial services, while business logic abuse—though less frequent—accounts for over 60% of financial losses from API incidents.

Traditional security approaches—including SAST, DAST, and WAFs—were found to be inadequate for addressing these critical vulnerability types. While these tools perform acceptably for injection vulnerabilities and misconfigurations, they fail to detect authorization vulnerabilities and business logic abuse. This detection gap is reflected in the case study findings, where all institutions experienced API incidents despite substantial investment in traditional security tools.

The research demonstrated the potential of machine learning and behavioral analytics to address these limitations. Anomaly detection, behavioral profiling, and sequence analysis can achieve detection rates exceeding 75% for business logic abuse, substantially outperforming traditional approaches. However, implementation challenges—including data requirements, false positives, and adversarial evasion—must be addressed for effective deployment.

Based on these findings, the research proposed a comprehensive detection framework incorporating six integrated layers: perimeter protection, static analysis, dynamic testing, anomaly detection, behavioral analysis, and threat intelligence integration. The framework is designed to address the limitations of individual approaches through complementary layers, each targeting different vulnerability types and attack patterns.

The research also contributed a cybersecurity resilience model specifically for open banking APIs, emphasizing detection, response, and recovery capabilities. The model includes maturity levels to guide institutions in progressively improving their API security capabilities.

6.2 Contributions

6.2.1 Theoretical Contributions

This research makes several theoretical contributions. First, it provides a comprehensive characterization of API vulnerabilities in open banking ecosystems, integrating findings from literature analysis, vulnerability trend analysis, and case study research. This characterization identifies the most critical vulnerability types and their patterns, informing theoretical understanding of API security in financial contexts.

Second, the research synthesizes findings on detection approaches—including traditional approaches, ML-based approaches, and integrated approaches—providing a framework for understanding their respective effectiveness and limitations in open banking contexts. This synthesis identifies research gaps and informs the development of more effective detection strategies.

Third, the research proposes a theoretical model for API resilience in open banking, incorporating detection, response, and recovery capabilities. This model provides a conceptual foundation for understanding and improving API security capabilities.

6.2.2 Practical Contributions

From a practical perspective, the research provides actionable guidance for financial institutions seeking to strengthen API security. The detection framework offers a concrete approach for implementing comprehensive API security, including both traditional and advanced detection capabilities. The resilience model provides a roadmap for progressive improvement, enabling institutions to assess their current capabilities and plan for enhancement.

The research also provides guidance on regulatory compliance, identifying areas where regulatory requirements align with security best practices and where tensions exist. This guidance helps institutions navigate regulatory obligations while maintaining effective security.

For policymakers and regulators, the research highlights considerations for open banking security regulation, including the need for advanced detection capabilities, harmonized requirements, and threat intelligence sharing.

6.3 Limitations

The research has several limitations that should be acknowledged.

Case Study Scope: The six case study institutions, while diverse, represent a limited sample of financial institutions. Findings may not be generalizable to all institutions, particularly smaller institutions with different resources and capabilities. The research acknowledges this limitation and discusses generalizability accordingly.

Data Availability: Vulnerability trend analysis is limited by the availability of data. Not all vulnerabilities are reported in the NVD or other databases, and not all incidents are publicly reported. The analysis may undercount vulnerabilities and incidents, particularly those that are discovered and remediated without public disclosure.

Self-Reporting Bias: Case study findings rely on self-reported information from institutions, which may be subject to bias. Institutions may underreport incidents or overstate security capabilities. Triangulation with secondary data and publicly reported incidents partially addresses this limitation.

Limited Generalizability: The framework and model proposed in this research are conceptual, developed from literature and case study findings. Empirical validation in operational environments is needed to establish effectiveness and refine implementation guidance.

6.4 Future Research Directions

The research suggests several directions for future investigation.

Empirical Evaluation of Detection Framework: The proposed detection framework should be evaluated in operational financial environments to validate its effectiveness and identify implementation challenges. Research should examine detection rates, false positive rates, operational performance, and integration requirements.

Business Logic Abuse Detection: Given its prevalence and impact, business logic abuse detection requires focused research. Sequence analysis, transaction modeling, and behavioral profiling approaches should be further developed and evaluated in financial contexts.

Adversarial Robustness: Research on adversarial robustness of ML-based detection is essential to address the threat of attackers evading detection systems. Studies should examine how attackers can evade detection and develop countermeasures.

Regulatory Security Tensions: Research is needed on the relationship between regulatory compliance and security outcomes, particularly where regulatory requirements create security tensions. This research could inform more effective regulation that improves security rather than creating vulnerabilities.

Cross-Jurisdictional Comparison: Comparison across open banking frameworks and jurisdictions could provide insights into regulatory effectiveness and identify best practices that can be adopted globally.

Automated Response: While this research focused primarily on detection, automated response capabilities are increasingly important. Research should examine automated response approaches for API incidents, including automated isolation, credential rotation, and notification.

6.5 Final Remarks

The open banking revolution promises significant benefits to consumers, businesses, and financial institutions—increased competition, innovation, and choice in financial services. However, these benefits depend critically on the security of the API infrastructure that enables open banking. If consumers do not trust that their financial data will be protected, the promise of open banking will remain unrealized.

This research provides insights, frameworks, and guidance for strengthening API security in open banking ecosystems. The findings emphasize the need for sophisticated detection capabilities that go beyond traditional approaches, incorporating ML-based analytics and behavioral context. The proposed detection framework and resilience model offer practical guidance for institutions seeking to improve their API security.

Ultimately, effective API security requires a combination of advanced detection, robust response, and continuous improvement. It requires investment in capabilities, skills, and processes. And it requires collaboration across institutions to share threat intelligence and best practices. While the challenges are significant, they are not insurmountable. With the right strategies and investments, financial institutions can secure the API infrastructure that powers open banking and realize the full potential of this transformative paradigm.

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